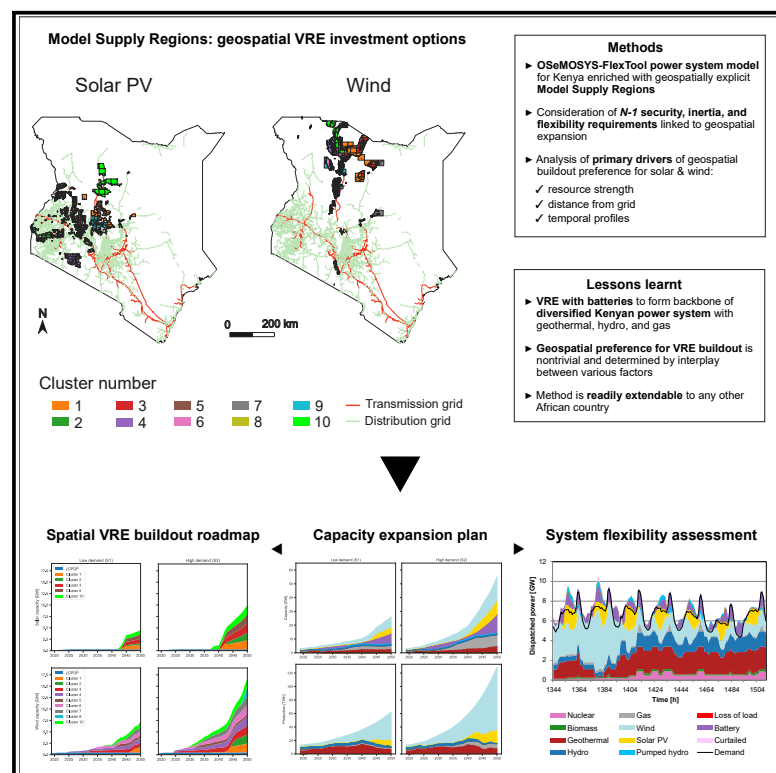


The map behind the roadmap—Introducing a geospatial energy model for utility-scale solar and wind power buildout in Kenya

Graphical abstract



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In brief

Power systems planning in African countries faces many challenges, including growing demand, weak grids, and the integration of variable renewables (wind and solar), and often neglects geospatial aspects. Millot et al. combine a geospatially explicit VRE potential model with a capacity expansion and dispatch model in Kenya to analyze the drivers behind the VRE buildout. Results show that site selection for VRE buildout is nontrivial and depends on various elements, including capacity factors, distance from the grid, and temporal profiles.

Highlights

- A geospatially explicit description of VRE enhances power system model planning
- The future Kenyan power system can rely on high penetration of VRE alongside batteries
- The VRE site selection is nontrivial and not solely determined by capacity factors



Article

The map behind the roadmap—Introducing a geospatial energy model for utility-scale solar and wind power buildout in Kenya

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SCIENCE FOR SOCIETY Advancements in electricity access, economic development, and population increase are just some of the elements driving electricity demand growth in Africa. Open and transparent energy system models, coupled with geographically detailed and reliable data, are essential tools for planning sustainable systems capable of providing low-cost electricity for all. This study helps address these challenges by introducing a methodology to identify the optimal locations for solar and wind power plants, considering the trade-off between exploiting the best resources and the costs of expanding the electrical grid to reach them. The method, readily applicable to all African countries, is showcased in Kenya, where solar and wind resources, coupled with batteries, could constitute the backbone of a diversified power system. The optimal selection of sites is driven by a nontrivial interaction between resource strength, distance from the grid, and temporal profiles, proving the added value of introducing explicit geospatial information in energy system models.

SUMMARY

For many African countries, energy models point toward variable renewable energy (VRE), mainly solar photovoltaics and wind power, as the potential backbone of future power systems. However, such models have not typically been able to explicitly include geospatial aspects around future VRE power plant siting with respect to cost optimization. For instance, should one build far from the grid in search of excellent resources or rather stay close to existing infrastructure to keep costs low? Here, we present an open-source energy modeling system (OSeMOSYS)-FlexTool workflow for capacity expansion and dispatch optimization planning for African countries that explicitly includes the geospatial dimension of VRE buildout. Applying the model to Kenya, we identify clear and nontrivial links between site characteristics and siting preferences in the optimization results, which moreover differ markedly for solar and wind power. This highly replicable approach has important implications for power system planning, especially in African countries whose grids will be VRE-heavy in the future.

INTRODUCTION

For a successful transition to sustainable power systems that enable access to low-cost electricity for all, a strong expansion of power generation capacity will be required across the world to meet rising demand. The main drivers of rising demand are (1) ongoing economic development alongside population growth, in particular in many developing nations,¹

session of power generation capacity will be required across the world to meet rising demand. The main drivers of rising demand are (1) ongoing economic development alongside population growth, in particular in many developing nations,¹



and (2) the required drive for end-use electrification in transport, buildings, and industry as nations embark on pathways toward decarbonization.²

To attain such long-term expansion of power systems, various stages of planning are required. For instance, the International Renewable Energy Agency (IRENA) recommends four subsequent stages^{3,4}:

- (1) long-term planning exercises (several decades into the future) to obtain scenarios on possible low-cost combinations of power-generating sources constituting an adequate power mix;
- (2) geospatial planning exercises to determine where to build the power plants constituting the aforementioned mix and to determine how and where to build out the grid;
- (3) dispatch studies (at hourly or sub-hourly timescales) to ascertain that the suggested power mix would be robust across variable situations of demand and supply availability; and
- (4) grid studies (at sub-second timescales) to study the static and dynamic stability of the grid under extreme situations and allow contingency planning.

Each of these stages typically requires its own dedicated modeling tools and methodologies. For instance, the academic literature has firmly established cost-optimization models as appropriate for the long-term planning of stage 1, and a substantial body of literature has developed around the use of models such as open-source energy modeling system (OSeMOSYS), LEAP, EnergyPLAN, SPLAT-MESSAGE, TIMES, and others that are now seen as standard. In general, there is an ongoing trend of moving to models with higher spatiotemporal resolution,^{5–8} which is necessary in the face of strong growth of variable renewable energy (VRE) sources in recent years. Simultaneously, there is another trend of moving toward more open-source solutions to allow for more straightforward use of these models by policy planners and practitioners.^{9,10}

Despite this growing body of work, various regions of the world still suffer from a lack of attention by the research community. In particular, the African continent is currently the least covered in terms of energy modeling studies at the country level,^{11,12} despite having both the world's fastest-growing demand and the world's highest need for energy access growth and grid buildout.^{13,14} Given the latter challenges, energy models—which were originally largely designed in and adopted for European and North American geographies, where the same challenges do not necessarily apply—typically require certain adaptations to be suitable for use in most countries in Africa.¹⁴

This divergence of model requirements between Europe/North America on one hand and Africa on the other becomes clear when considering the combination of stage 1 above (long-term planning) with stage 2 (geospatial planning) and stage 3 (dispatch planning). Energy modeling exercises for countries across Africa, as for many other parts of the world, nowadays tend to point toward solar photovoltaics (PVs) and (depending on geography) wind power as the most promising future technologies to meet growing power demand, supported by hydropower and geothermal where available, alongside a strong drive

toward grid strengthening, interconnections, and storage solutions.^{1,15–20} But, as opposed to other parts of the world, many African countries are not currently in a position to implement the solar and wind power buildout that such long-term planning models may suggest, as they face continued challenges with relatively weak grids with low levels of inertia, low spatial buildout of grids covering only limited parts of national territories, and high costs of capital for VRE technologies^{13,14} as starting points. These challenges impact all stages mentioned above, from long-term planning to short-term planning to geospatial planning. Thus, the classical method in which these are separate and subsequent planning exercises feeding into each other is likely not sufficient in the context of many African countries and would lead to sub-optimal planning. This could be avoided by unifying them as much as possible within one single approach.^{13,14}

This implies that long-term planning—which tends to suggest high future shares of VRE—should, *from the outset*, account for geospatial planning (Where should the suggested VRE be built, taking into account constraints dictated by, e.g., land access/use, terrain, and distance from existing infrastructure? What makes some locations more attractive than others at the system level?), dispatch planning (How should the rest of the system operate as VRE keeps growing? Where will flexibility come from? How can sufficient inertia in the growing system be ensured at all times?), and grid expansion planning (Are there enough transmission capacities between regions? How can *N-1* security be ensured in the growing system?). The geospatial elements of VRE buildout and concurrent grid expansion are evidently crucial in unifying these different planning elements.

There are existing modeling frameworks that include geospatial aspects in optimization models, and these have been applied in valuable studies, focusing on, e.g., Europe,^{21,22} the United States,²³ and China,^{24,25} some of which have, more recently, been expanded to Africa.^{26,27} However, these studies have typically either had quite different objectives or tended to leave out at least one of the above-mentioned elements of unification between capacity expansion planning, geospatial planning, and dispatch planning. For instance, in Hörsch and Brown²¹ and Kies et al.²² on the European power system, the focus was on better representing transmission bottlenecks in multi-node models as compared with one-node-per-country models. In Hörsch and Brown²¹ and Kies et al.²² on geospatially resolved grid expansion scenarios for Germany, the focus was on simulation and not optimization, and the potential locations of future transmission lines were pre-defined, with no specific grid buildout costs allocated to individual VRE projects within each node. Further, although site-specific VRE investment options are part of the optimization model used in Patankar et al.,²³ the study concerning the United States focused on potential conflicts between cost-optimal site selection and land use patterns and did not report on the specific drivers behind selection of individual sites. Similarly, for China, Zhuo et al.²⁴ used geographically resolved data with distance-dependent costs for VRE as the model input but aggregated this information at a province level for running the model, giving one node per administrative boundary and not one node per area with similar VRE resources, thus reducing the potential of model results to elucidate the best locations to deploy VRE. And, although the PyPSA modeling

framework^{26–28} allows the inclusion of rich spatial detail on VRE potential, existing PyPSA applications did not explicitly assess the drivers of geospatial site selection nor the consequences of geospatial renewables buildout on meeting *N-1* security and inertia needs.^{26–28} Moreover, the latter studies typically differentiated between VRE areas on the basis of resource potential alone (if not starting from the basis of only allowing VRE expansion along existing transmission corridors to start with^{26,27}), in both cases precluding an analysis of the effect of remoteness from existing infrastructure, or the effect of discrepancies between temporal profiles across sites, on VRE buildout preferences.^{26,27} Further, such studies have tended to focus on single years in the future,^{26–28} without considering the capacity expansion timeline over the years and the preferred order of geospatial buildout for cost optimality. This is also notably the case for the recently published study by Zhang et al.,²⁵ which uses highly detailed, geospatially resolved data for a model of China's power system, simulating the year 2060 and constraining the model to attain carbon neutrality, but does not analyze timelines of selecting sites. In fact, the timelines of geospatial capacity expansion are particularly relevant for countries where the power system is expected to grow substantially over the next decades in the quest for higher levels of electricity access, such as most African countries.^{13,14} In that context, although recent progress has been made to improve geospatial representation in energy models for Uganda²⁹ and Kenya,³⁰ the main focus of those studies was modeling on-grid and off-grid expansion within the same framework, again without considering the geospatial element of utility-scale VRE expansion in optimized long-term and dispatch planning.

Thus, although substantial progress has been made to include geospatial aspects in energy system models, a fully unified framework integrating geospatially differentiated VRE investment options with both long-term planning and dispatch planning, including grid security and stability considerations from the outset, is still an area of research in energy modeling literature, with various open questions on, e.g., the potential drivers of site selection. For countries and regions where such unified model frameworks already exist, such as the United States and China, the underlying drivers of site selection in respect of cost optimization have not yet been reported in detail, whereas for many other countries, especially on the African continent, the frameworks do not yet exist. In the latter case, this can lead to possible discrepancies at the African country level between contemporary long-term visions (typically, strong VRE expansions) and on-the-ground spatial power sector planning, often characterized by hesitation to prioritize VRE, both due to grid stability concerns in the context of relative inadequacy of existing grid infrastructure and due to the absence of grid infrastructure in locations with attractive VRE resources.^{13,14}

The above-mentioned challenges have so far prevented the design of realistic, spatially explicit, priority roadmaps to attract investment for VRE plants and concurrent masterplans for grid expansion across the African continent. Elsewhere, such roadmaps are already common, as seen from, e.g., the Dutch government's recently published spatial offshore wind roadmap.³¹

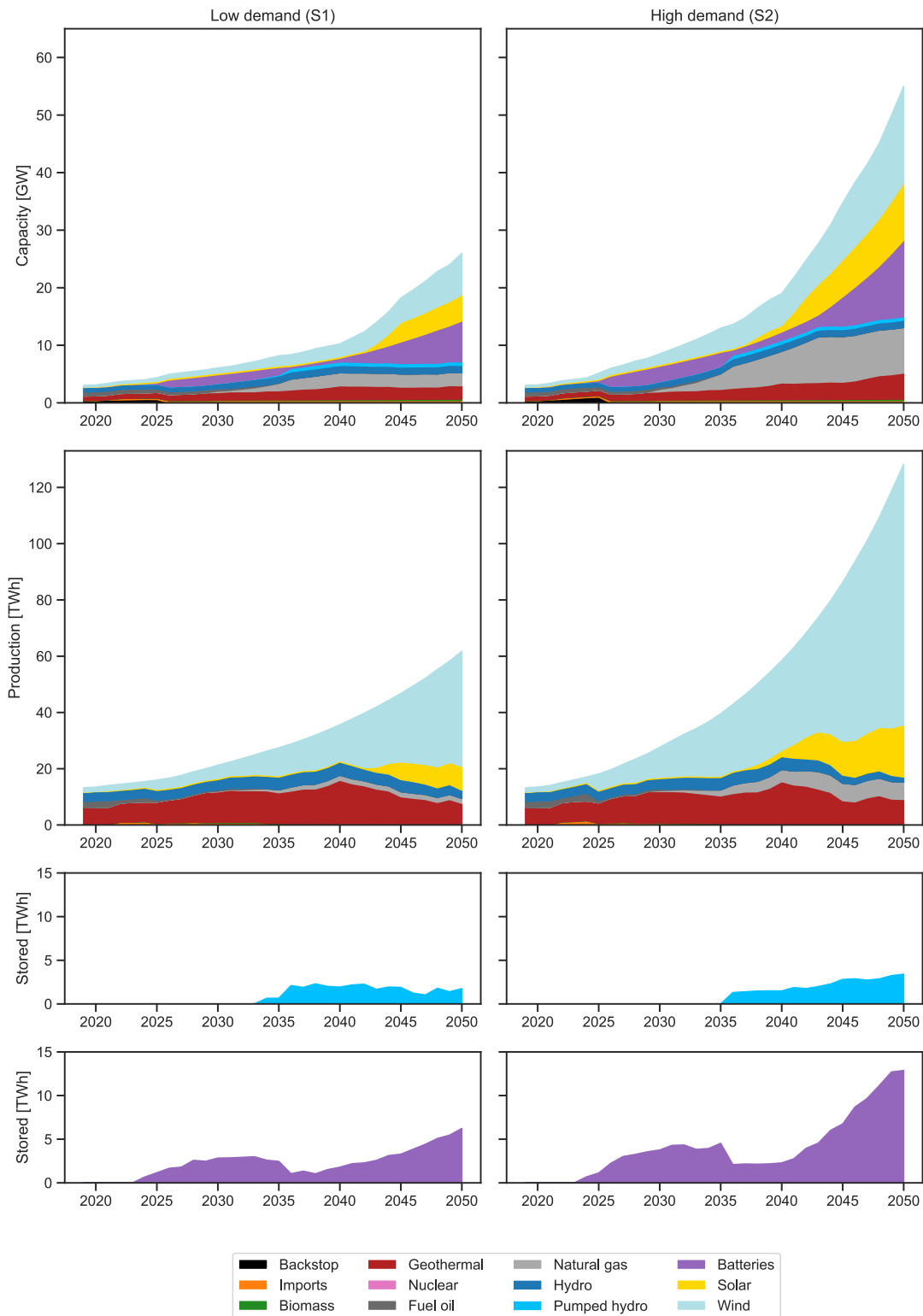
Approach

In this study, we present a new method, adapted specifically for countries with high-demand growth, relatively weak existing grids, and potentially data-scarce environments (such as many countries on the African continent), to combine capacity expansion planning, geospatial power system planning, and dispatch planning into a single coherent approach, essentially merging the main elements of stages (1)–(3) above into a single simplified workflow. We demonstrate this approach through an OSeMOSYS-FlexTool workflow for Kenya, where recent work has suggested solar and wind power are set to become the dominant power-generating sources by 2050.³² For details on this workflow, the reader is referred to “experimental procedures,” sub-section “OSeMOSYS-FlexTool linkage.”

We chose Kenya as a case study for two reasons. First, the potential for both solar PVs and wind power in Kenya is high, and both tend to emerge as “backbone” sources for Kenya's grid, according to classical capacity expansion studies,³² in line with Kenya's commitments toward decarbonizing.³³ Second, Kenya faces ongoing challenges in its power sector, in particular the challenge of blackouts, which have recently been linked to insufficient holistic planning in the wake of increasing VRE penetration, with, e.g., loss of generation at the Lake Turkana Wind Power Plant having been blamed for nationwide outages.^{34,35} This indicates an imminent need for a comprehensive method integrating capacity expansion planning, geospatial planning, and dispatch planning.

For this study, we upgraded an existing OSeMOSYS-FlexTool workflow for Kenya (see Kihara et al.³² for details) by including spatially explicit information on potential VRE locations—including location-dependent resource strength and temporal power generation profiles, site-specific transmission grid and road network expansion costs, and terrain-constrained available land area for each location—in the optimization procedure, according to the “model supply regions” (MSRs) approach developed by IRENA.³⁶ Essentially, instead of using a single representative technology for solar PV and wind power each among the investment options in the capacity expansion model, we split up the investment options for solar and wind in the model into spatially disaggregated clusters—ten for solar and ten for wind—each with its own characteristic power generation profile, its own characteristic transmission grid and road network expansion costs to link it to existing infrastructures, and its own specific maximum deployment potential according to constraints dictated by the terrain. For details, the reader is referred to “experimental procedures,” sub-section “representation of solar and wind investment options.”

Subsequently, by explicitly linking *N-1* system security constraints³⁷ and inertia constraints³² to the spatially divergent VRE expansion options both in OSeMOSYS and FlexTool, the workflow integrated long-term capacity expansion scenarios on one hand, with hourly scale dispatch scenarios for individual years on the other (again, see “experimental procedures,” sub-section “representation of solar and wind investment options” for details.) This allowed us to produce coherent scenarios merging the short-term, long-term, and geospatial elements of planning. In this way, credible geospatial investment roadmaps for



(legend on next page)

utility-scale solar and wind power expansion for Kenya could be drawn up.

This is, to our knowledge, the first such application of geospatially explicit information in a coupled long-term capacity expansion and short-term dispatch workflow for an African country context. Although Kenya is used here as a case study, given that the approach and implementation are fully based on open-source workflows and datasets, the same principle could be readily applied to any other country in Africa or other nations sharing some of the typical characteristics of African power systems.¹⁴ We note that the intended novelty of our study lies not in the application of geospatially explicit datasets on VRE as such, which various studies have done before (see [introduction](#)), but rather in the conception and application of a geospatial referencing framework suitable for typical African country contexts that tend to share certain characteristics currently not present in most other parts of the world.¹⁴

Most of the assumptions have been chosen in collaboration with the Least Cost Power Development Plan (LCPDP) team of the Kenyan Ministry of Energy and Petroleum, which produces the medium- and long-term plan focusing on generation and transmission for the Kenyan power system.^{38,39} One of the authors of this paper (M.A.) was part of the LCPDP team during the conception of this paper.

RESULTS

Power system capacities in OSeMOSYS

The evolution of the Kenyan power sector toward 2050 for two demand scenarios modeled with OSeMOSYS is shown in [Figure 1](#). Both demand projections are obtained from Kihara et al.³²; the low-demand scenario (denoted with S1) is based on historical growth rates and results in an average year-on-year growth rate of 5.3%, whereas the high-demand scenario (S2) is consistent with all targets of the Vision 2030 development program being met,⁴⁰ projecting the same average annual growth of 8.2% until 2050.

Three phases can be identified in the evolution of the power system.

The first few years are reflective of the current situation in Kenya, characterized by the occurrence of blackouts partly caused by insufficient capacity. Indeed, the capacity available to the model between 2019 and 2025, constrained by existing capacity and committed projects as per Kenya's medium-term plan,^{38,39} is not enough to meet the capacity requirements of the system, and OSeMOSYS is forced to install some fictitious backstop capacity (see black wedge in graph). However, from 2026, the capacity buildout constraints in the model are relaxed, and the model is free to install enough new capacity to meet the extra power needs. The bigger contribution to the early expansion of the system is provided by batteries, which provide two distinct services to the grid. First, batteries help meet the reserve margin needs. Based on the LCPDP team planning procedures,

a 10% reserve margin is imposed across the model time horizon, meaning that the total installed firm capacity has to exceed by 10% the annual peak power demand. Batteries are allowed to contribute to the fulfillment of the reserve margin and are hence invested in by the model to solve the grid issues as soon as they become available (see [Table S1](#) for an overview of all technologies' reserve margin contributions). The second way in which batteries contribute to the expansion of the system is by redispatching the energy produced by VRE—mostly wind turbines at this point of the model time horizon. The amount of energy stored and then redistributed by batteries is shown in purple in the lower panel of [Figure 1](#). In both demand scenarios, battery activity surges around 2025 to reach a first peak in activity between 2030 and 2035.

Natural gas power plants are not permitted to come online before 2030 in the model, as Kenya is currently lacking any type of gas infrastructure. New capacity is considered to be based on liquefied natural gas (LNG) imports, and the way in which the relative costs are considered is outlined in “[experimental procedures](#),” subsection “[OSeMOSYS description](#).” From 2030 onward, natural gas capacity starts to be added, progressively replacing the first wave of battery installations as they are retired. As was the case for batteries in the 2020s, gas power plants are mostly built to satisfy the reserve margin requirements, and this can be seen by comparing the share of natural gas in installed capacity (upper panel in [Figure 1](#)) to the share of natural gas in electricity generation (second row panel in [Figure 1](#)). In the former case, natural gas reaches a peak share of 23% in 2038 in the low-demand case and 29% in the high-demand one on a capacity basis, whereas the electricity production peaks from natural gas plants are limited to 5.3% and 7.8%, respectively, showing that the added natural gas capacities are idle for most of the year and used exclusively as peaking plants. The electricity generation mix from 2030 to 2040 is mostly composed of geothermal, hydro, and wind power.

The third and last phase of the evolution of the Kenyan power system starts around 2040, when the combination of solar PV and batteries becomes cost-competitive with geothermal and gas power plants. Most additional capacity requirements after 2040 are met by solar, wind, and batteries, reaching a total installed capacity of 26 GW (low demand) and 55 GW (high demand) in 2050, a manifold increase from the 2.7 GW in 2019.

Other minor trends that can be observed include the low reliance on imports from Ethiopia throughout the entire modeled period, despite 400 MW being available from 2026; the modest contribution from pumped hydro, limited by the 600-MW technical potential currently estimated by the LCPDP team; and fuel oil plants being completely retired in all scenarios before 2035.

In addition to these two scenarios, two additional scenarios were run in which the cost outlook for solar and wind power was somewhat less optimistic, to show the robustness of the obtained results even under potentially more-adverse economic

Figure 1. Power system capacities and levels of activity in OSeMOSYS

OSeMOSYS results for the low (left column) and high (right column) demand scenarios. The first row shows the evolution of the installed capacity throughout the modeled time horizon, whereas the second row does the same for the actual energy produced by each power source. Rows three and four show the amount of energy stored and discharged each year by, respectively, pumped hydro and batteries. See also [Figure S1](#) for additional scenarios.

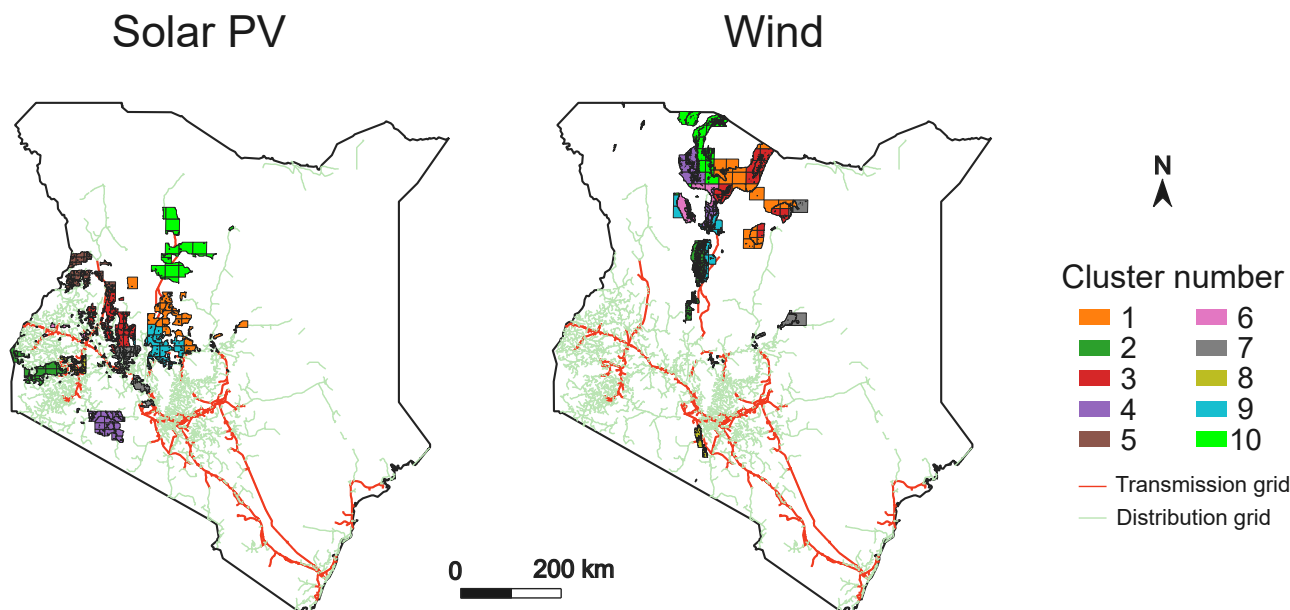


Figure 2. Geospatial representation of the solar and wind clusters for Kenya

The figure shows the clusters for solar PV (left) and wind power (right) included in the present OSeMOSYS-FlexTool workflow. The data on clusters are taken from the repository in Sterl et al.⁴¹ and re-visualized for purposes of this paper. The data on the transmission and distribution grid are taken from GridFinder.⁴² Note that the numbers of the clusters are arbitrarily assigned and do not signify any particular order of attractiveness. Both sets of clusters cover 5% of Kenya's territory, and the total potential for solar PV and wind power plant deployment across clusters adds up to 96 GW (solar) and 86 GW (wind).

conditions for VRE. More information is provided in “[experimental procedures](#),” sub-section “[scenarios](#),” and results are shown in the [supplemental information](#) (Figure S1).

Geospatial analysis of results

Having described the overall makeup of Kenya's capacity mix according to the OSeMOSYS results, we can now analyze the selection of solar and wind power clusters in greater detail, as well as the dynamics behind the model's preference for their buildout. We note that the buildout speed of VRE within each cluster is constrained according to $N-1$ security criteria (see “[experimental procedures](#),” subsection “[representation of solar and wind investment options](#)”), which prevents the model from concentrating all capacity in single locations with highly attractive conditions. This constraint therefore pushes the model to deploy VRE across multiple locations according to a priority ranking of clusters, typically starting with a single preferred region and progressively spreading across more and more clusters. A geospatial representation of the clusters included in the model workflow, adapted from Sterl et al.,^{36,41} is provided in Figure 2.

Our aim here is to analyze the characteristics of this priority ranking and how it differs for solar and wind. In particular, we assess the added value of the geospatial representation of solar and wind investment options through the MSR clusters as compared with classical energy modeling approaches that lack this geospatial dimension (or that add it in a post-processing step, not nested in the optimization). Such an added value becomes evident if the model selects solar and wind clusters in a *nontrivial* order, that is, if the selection preference for solar and

wind clusters does not simply follow whichever cluster has the highest yield per unit of capacity (expressed through the average capacity factor [CF]).

In general, we can observe that this approach provides three distinguishing features of the individual clusters, which is distinct from the standard OSeMOSYS approach: (1) the average resource strength (CF), (2) the supplementary costs for transmission grid and road buildout ($trCAPEX$), and (3) the temporal profile of the resource. Thus, if the model prefers selecting clusters in a given order, this order must be interpretable by examining the aforementioned three features.

The buildout of MSR clusters is shown visually in Figure 3, which provides a breakdown by cluster of the overall solar and wind buildout across Kenya already shown in Figure 1. In Tables 1 and 2, we now show a detailed overview of the characteristics of all the solar and wind clusters selected by the model runs, in order of selection by the model. The tables include details on the buildout, notably their first year of operation and total capacity by 2050 according to OSeMOSYS results. Further, we show each cluster's average CF and average supplementary CAPEX for grid and road buildout ($trCAPEX$) and for wind power, as well as the wind turbine class allocated to each specific cluster as per Sterl et al.^{36,41} We note that CF and $trCAPEX$ can be combined into a single indicator, namely the levelized cost of electricity (LCOE), including the infrastructure expansion CAPEX. For the sake of better understanding the individual drivers and trade-offs of geospatial selection, we show CF and $trCAPEX$ separately in the tables; however, graphical representations of the LCOE-based marginal cost curves of solar and wind clusters are given in Figures S3 and S4. Last, Tables 1

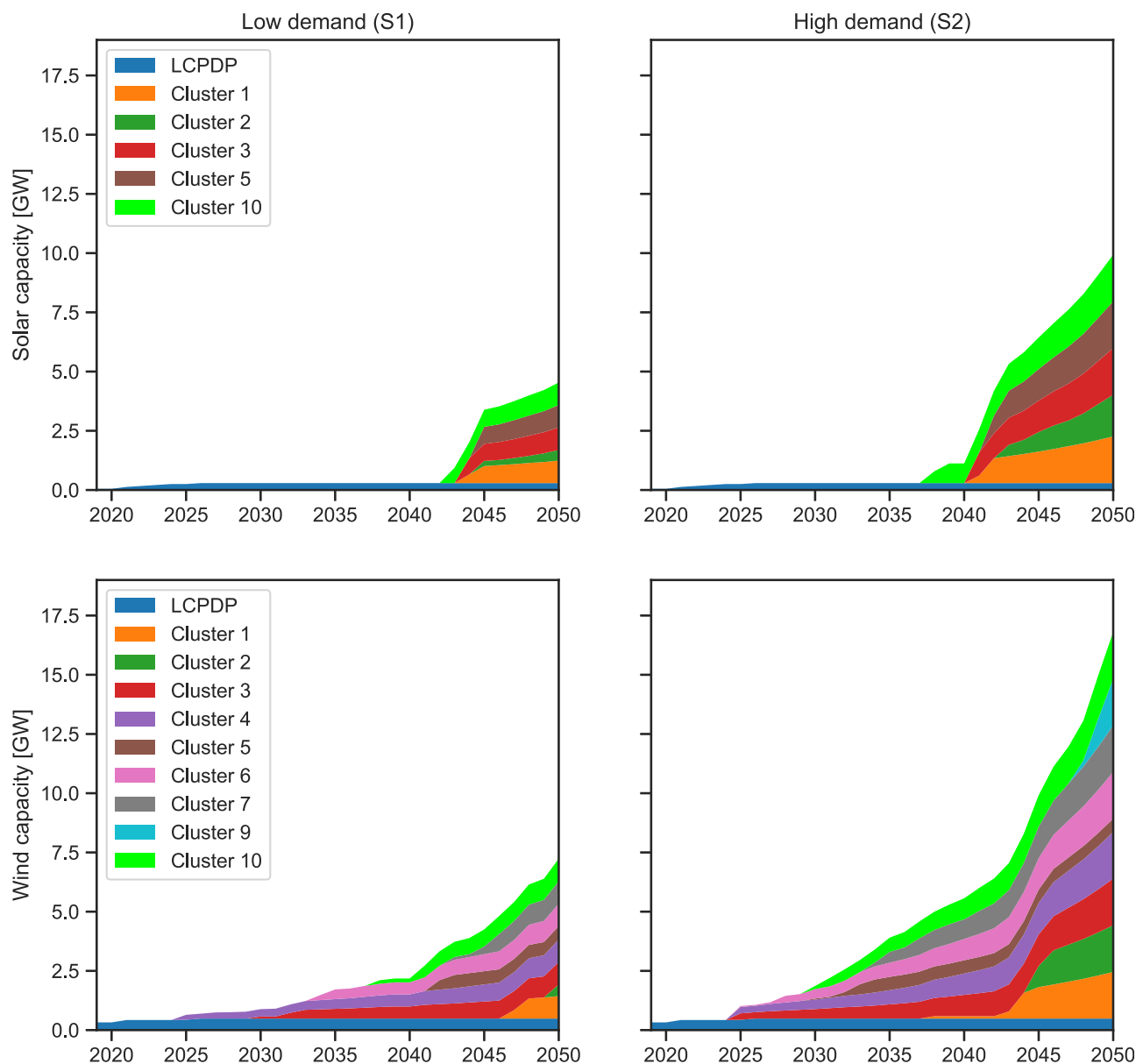


Figure 3. Capacity buildout of the geospatially referenced solar and wind clusters

MSR cluster buildout according to OSeMOSYS for the low (left) and high (right) demand scenarios. The first row shows the solar MSR cluster buildout and the second row the wind MSR cluster buildout. See also [Figure S2](#) for additional scenarios.

and [2](#) include a qualitative description of the geographical characteristics and the lay-of-the-land of each cluster.

This overview allows for an in-depth assessment of why certain locations are preferred to others. Looking first at solar PV ([Table 1](#)), it is clear that the selection order does not trivially follow the resource strength and that distance from infrastructure plays an important role. For instance, although the highest *CF* solar cluster (#10, with *CF* = 21.4%) is selected first, the second-highest-*CF* solar cluster (#4, with *CF* = 21.2) is never selected, which can be explained by its remoteness and, therefore, prohibitively high *trCAPEX*, translating into an unfavorable LCOE despite having the second-best resource quality (cf. [Figure S3](#)).

The decisiveness of *trCAPEX* in certain cases is observed most clearly among selected solar clusters with highly similar *CF*, such as solar clusters #2, #3, and #5, which all exhibit a *CF* = 20.8%. Here, the order of selection appears to be determined by the *trCAPEX* parameter, with solar cluster #3 being selected earlier than solar cluster #5 (which has a 5% higher *trCAPEX*) and solar cluster #2 (which has a 15% higher *trCAPEX*). In other words, because there is nothing to be gained in *average* yield by changing between solar clusters #2, #3, and #5, and because the temporal profiles of these clusters have relatively similar seasonal profile shapes (with all three clusters lying in the westernmost part of Kenya, cf. [Figure 2](#)), the

Table 1. Solar PV clusters						
Cluster # (maximum capacity in GW)	First year	Capacity in 2050 (GW)	CF (%)	trCAPEX (USD/kW)	County	Qualitative description
Selected in all scenarios						
#10 (18.93)	2043 (S1); 2038 (S2)	0.95 (S1)*; 1.97 (S2)*	21.4%	141.6	Marsabit, Samburu	areas on the eastern side of the Rift Valley directly to the south of Lake Turkana; this cluster is partially coincident with wind clusters #4 and #9
#1 (15.86)	2044 (S1); 2041 (S2)	0.95 (S1)*; 1.97 (S2)*	21.0%	120.6	Laikipia, Samburu	area on the highlands bordered to the north by Maralal town and the Loroghi Plateau, to the west by the Rift escarpment, and to the east by the western slopes of the Milgis river sub-basin
#3 (12.74)	2044 (S1); 2041 (S2)	0.95 (S1)*; 1.97 (S2)*	20.8%	122.1	Baringo, Elgeyo-Marakwet	lower-lying areas of the Rift Valley as well as in the adjacent Kerio Valley, bounded by the B4 highway in the north and Lake Bogoria in the south
#5 (8.69)	2045 (S1); 2042 (S2)	0.95 (S1)*; 1.97 (S2)*	20.8%	128.5	West Pokot, Trans-Nzoia	highland area along the Kenyan-Ugandan border roughly corresponding to the triangle between Mount Elgon in the southeast, Turkwell Dam in the north, and the Cherangani Hills in the southwest
#2 (7.09)	2045 (S1); 2043 (S2)	0.45 (S1); 1.76 (S2)	20.8%	140.2	Kisumu, Homa Bay, Siaya, Busia	lakeshore areas west and north of Kisumu straddling the shores of the shallow Winam Gulf; the authors note that there may be potential for deploying floating solar instead of ground-based solar here

(Continued on next page)

Table 1. Continued						
Cluster # (maximum capacity in GW)	First year	Capacity in 2050 (GW)	CF (%)	trCAPEX (USD/kW)	County	Qualitative description
Not selected						
#4 (9.48)	–	–	21.2%	165.6	Narok	area on the plateau directly southwest of Narok town, part of the Southern Ewaso Ng'iro basin, bounded in the south by the Maasai Mara National Reserve
#6 (1.64)	–	–	20.6%	103.0	Uasin Gishu, Nandi	areas on the plateau directly surrounding Eldoret town west of the Rift Valley
#7 (7.32)	–	–	20.6%	107.4	Nakuru, Baringo	southward continuation of cluster #3, comprising low-lying areas in the Rift Valley north and south of Nakuru, roughly stretching from Lake Bogoria to Lake Naivasha
#8 (2.73)	–	–	20.3%	88.3	Nandi, Kericho, Kisumu	the low-lying area stretching eastward from Kisumu (Winam Gulf of Lake Victoria) up to the escarpment on the western side of the Rift Valley, sandwiched between higher-lying areas to the north (Nandi Hills) and south (Mau Forest)
#9 (11.11)	–	–	20.8%	104.6	Nyeri, Laikipia, Nyandarua	similar to cluster #1 but further to the south, comprising areas in the highlands north of Mount Kenya and Aberdare National Parks; coincident to some extent with wind cluster #5, in particular around OI Pejeta Conservancy

Detailed description of the buildout timeline and distinguishing characteristics of the solar PV clusters in the model, ordered by their priority ranking as per the model's selection preference. A single asterisk (*) in the capacity column indicates that the capacity buildout by 2050 is constrained by *N-1* criteria, whereas a double asterisk (**) indicates that the capacity buildout by 2050 is constrained by the size of the cluster, and its potential is exhausted. Absence of an asterisk therefore indicates that buildout in the cluster has neither yet reached the maximum allowed by system security constraints nor the maximum spatial deployment potential within the cluster. S1, scenario 1 (low demand); S2, scenario 2 (high demand).

Table 2. Wind clusters

Cluster # (maximum capacity in GW)	First year	Capacity in 2050 (MW)	CF (%) and turbine class	trCAPEX (USD/kW)	County	Qualitative description
Selected in all scenarios						
#4 (10.07)	2025 (S1); 2025 (S2)	0.95 (S1)*; 1.97 (S2)*	55.4% (class-I)	245.4	Marsabit	areas directly straddling Lake Turkana on its eastern shore: the area south of Sibilo National Park and the area stretching from Loiyangalani town to the lake's southern tip; note that Lake Turkana Wind Power Plant is located in the latter zone
#3 (12.61)	2030 (S1); 2025 (S2)	0.95 (S1)*; 1.97 (S2)*	52.0% (class-I)	285.1	Marsabit	higher-lying areas surrounding Marsabit town, typically the lower slopes of mountains and hill ridges dotted across the area around (but not including) the Chalbi Desert; wind speeds are somewhat higher at these altitudes than in cluster #1, but the terrain is more difficult
#6 (5.74)	2034 (S1); 2025 (S2)	0.95 (S1)*; 1.97 (S2)*	50.0% (class-I)	218.5	Turkana, Marsabit	the area in Turkana County between Lake Turkana's southwestern shore and the Kerio River, as well as a smaller area on the opposite lakeshore, on the ridges north of Loiyangalani town in Marsabit County, where the lake's eastern shore curves sharply westward and where the shoreline itself lies in the ridges' wind shadow
#10 (15.62)	2038 (S1); 2030 (S2)	0.95 (S1)*; 1.97 (S2)*	45.1% (class-I)	334.8	Marsabit	area north of clusters #1, #3, and #4 on the eastern shore of Lake Turkana, stretching up to the Ethiopian border; this area is characterized by high wind speeds but much higher remoteness as compared with the other clusters in Marsabit county
#5 (0.56)	2042 (S1); 2030 (S2)	0.56 (S1)**; 0.56 (S2)**	40.7% (class-II)	107.0	Laikipa, Meru	areas around Mount Kenya National Park, in particular west of Mount Kenya straddling the Ol Pejeta Wildlife conservancy (a slight depression in the highlands) and north of Meru town; coincides to some extent with solar cluster #9
#7 (5.99)	2043 (S1); 2034 (S2)	0.95 (S1)*; 1.97 (S2)*	41.2% (class-II)	237.7	Isiolo	areas directly north of the Nyambene hill ridge and south of the Ewaso Nyiro river, where the highlands start their drop toward the Indian Ocean (there is an area with similar conditions at nearly identical longitude but more north in Marsabit County, where a similar drop toward sea level begins, but this area is harder to access)

(Continued on next page)

Table 2. Continued

Cluster # (maximum capacity in GW)	First year	Capacity in 2050 (MW)	CF (%) and turbine class	trCAPEX (USD/kW)	County	Qualitative description
#1 (20.37)	2047 (S1); 2038 (S2)	0.95 (S1)*; 1.97 (S2)*	41.8% (class-I)	245.3	Marsabit	lower-lying areas largely stretching between the hill ridges surrounding Marsabit town (which make up cluster #3), in particular the stretch from Marsabit town to North Horr town just north of the Chalbi Desert
#2 (6.05)	2050 (S1); 2045 (S2)	0.49 (S1); 1.97 (S2)*	46.6% (class-II)	191.2	Samburu, Turkana	areas mostly adjacent to and directly west and south of cluster #9, along the transition of the Losiolo escarpment into the Suguta Valley, where katabatic winds result in a higher average wind speed than cluster #9 but the locations are more scattered and harder to access
Additionally selected in high-demand scenario						
#9 (7.28)	2048 (S2)	1.97 (S2)*	42.4% (class-II)	148.6	Samburu	area just east of the Losiolo escarpment on the east side of the Rift Valley, roughly ranging in a north-south line from the southern tip of Lake Turkana to Malaso, interrupted by Mount Ng'iro, with the Suguta Valley to the west
Not selected						
#8 (1.64)	–	–	40.5% (class-III)	103.4	Kajiado	area running north-south, straddling the escarpment on the east side of the Rift Valley, starting around the Ngong Hills down to the latitude of Lake Magadi down in the Rift Valley; note that Ngong Hills wind power station is located here

Detailed description of the buildout timeline and distinguishing characteristics of the wind clusters in the model, ordered by their priority ranking as per the model's selection preference. A single asterisk (*) in the capacity column indicates that the capacity buildout by 2050 is constrained by *N-1* criteria, whereas a double asterisk (**) indicates that the capacity buildout by 2050 is constrained by the size of the cluster and its potential is exhausted. Absence of an asterisk therefore indicates that buildout in the cluster has neither yet reached the maximum allowed by system security constraints nor the maximum spatial deployment potential within the cluster. S1, scenario 1 (low demand); S2, scenario 2 (high demand). Note: wind turbine classes are allocated based on the average wind speed in each cluster, with class-I appropriate for average wind speeds > 8.5 m/s, class-II for average wind speeds between 7.5 and 8.5 m/s, and class-III for wind speeds below 7.5 m/s; the CF of each cluster is then determined by each turbine class's particular power curve.

determining factor for cluster preference becomes their distance from the existing infrastructure (and the associated additional cost for grid and road expansion), with locations closer to existing assets having lower $trCAPEX$ and thus lower LCOE (cf. [Figure S3](#)), being preferred.

Yet $trCAPEX$ (and, by extension, the LCOE) is not always the determining factor. For instance, we see in [Table 1](#) that solar cluster #9 also has the same $CF = 20.8\%$, as well as a lower $trCAPEX$ than the three aforementioned solar clusters (14% lower than solar cluster #3), resulting in the third-most attractive LCOE (cf. [Figure S3](#)), yet is still not selected. It is here that we observe that the shape of the profiles also plays a role in the selection process. To explain the non-selection of solar cluster #9, we need to draw into the discussion solar cluster #1, which is geographically contiguous to #9 ([Figure 2](#)). In this case, we observe that solar cluster #1 has a higher CF than solar cluster #9, which makes up for its somewhat higher $trCAPEX$ and gives it a marginally lower LCOE, aside from which its temporal profile is nearly indistinguishable from #9 due to their close geographical proximity. Together, these factors result in a marginally higher overall economic attractiveness to build in solar cluster #1 as compared with solar cluster #9. Once substantial capacity exists in solar cluster #1, there is no reason for the model to build in solar cluster #9, which has no real distinguishing features from #1 except for being marginally more expensive per unit of produced electricity. Solar clusters #2, #3, and #5, on the other hand, although exhibiting the same average resource strength as solar cluster #9 and being more distant from the grid, lie further to the west than #1 and #9 and, therefore, produce power until somewhat later in the evening, providing an added value in terms of diversifying the mix by correlating better with evening demand than solar cluster #9, despite having a higher LCOE. In summary, the model builds in solar cluster #1 (owing to its high CF) and solar clusters #2, #3, and #5 (owing to their correlation with demand and in an order determined by $trCAPEX$), but refrains from building in solar cluster #9 (which is slightly below solar cluster #1 in strength and not as synergetic to demand as solar clusters #2, #3, and #5).

Thus, it is shown that the selection preference for solar PV clusters is *nontrivial* as it does not follow the resource yield solely, and all three distinguishing features mentioned earlier (CF , $trCAPEX$, and temporal profiles) play a role. The added value of the geospatially referenced representation of solar PV clusters to find cost-optimal solutions is therefore evident.

Next, we turn to the wind results ([Table 2](#)). The higher geospatial variation of wind power potential as compared with solar PV across Kenya's territory means that the distance from existing infrastructures is less of an impediment to the attractiveness of developing wind sites vis-à-vis solar PV.³⁶ In other words, expanding existing infrastructures at high cost to connect a wind "goldmine" in a remote region may be more worth the effort as compared with solar PV. This is not only because CF differs much more between wind clusters than between solar clusters, dampening the relative effect of remoteness, but also because high wind CF values allow the use of class-I wind turbines, which have lower CAPEX than class-II and class-III turbines.^{36,41} This is evidenced by, e.g., the three wind clusters with the highest CF (all appropriate for class-I turbines) coming in first, second,

and third place in the selection order, in a sequence aligned with CF and seemingly unaffected by $trCAPEX$. Incidentally, cluster #4 (the first selected wind cluster) is the location of Kenya's existing Lake Turkana Wind Power Project, Africa's largest wind power plant, providing something of a real-life reflection of the preceding statements.

Nevertheless, the relationship between wind cluster selection order and wind power CF is not trivial. For instance, the small wind cluster #5 comes fifth in the selection order, despite having the second-lowest CF of all. Its selection before other clusters with higher CF could appear to be linked to its $trCAPEX$, which is second-lowest among all wind clusters and thus partially compensates the low CF . Yet, cluster #5 still only comes in third-last place in the marginal cost curve ([Figure S4](#)).

Analyzing the temporal profiles of the clusters, we observe that the CF time series of wind cluster #5 exhibits the highest correlation coefficient with load (at the time slice level) of all clusters, namely at a value of 0.39, with, e.g., the adjacent clusters #9 and #7 in the marginal cost curve only obtaining 0.14 and 0.05, respectively (note that the same trend is also true at full hourly level). The comparatively good correlation with demand of wind cluster #5 thus translates into lower requirements for additional investment in flexibility to match supply and demand and makes wind cluster #5 attractive to invest in, despite its small size and notwithstanding the apparent lower attractiveness of other wind clusters in the same class with comparable or better LCOE, like wind cluster #9.

A similar argument can be made for the competition between wind clusters #10 and #2. Discrepancies in CF , $trCAPEX$, and turbine class CAPEX between the two partially cancel out, and, as a result, the two clusters are adjacent in the marginal cost curve ([Figure S4](#)), yet cluster #10, the more expensive of the two, is selected more than a decade earlier than #2. Despite its less-favorable LCOE, wind cluster #10 has a more constant resource throughout the year than #2 (coefficient of variation of 0.22 for the former's time-sliced profile, as compared with 0.29 for the latter) and, perhaps more importantly, #10 exhibits exquisite seasonal synergy with demand (with a correlation coefficient of 0.95 between the seasonal profiles of cluster #10 and demand as compared with only 0.55 for #2, whose resource has a much more unfavorable seasonality). Therefore, building in wind cluster #10 is preferable as it requires less investment in additional flexibility and seasonal balancing as compared with wind cluster #2 (although in the end the latter is also built).

We note here that even though all selected wind clusters are at their maximum allowed capacity by 2050 in the high-demand scenario, the model still does not select wind cluster #8. This is because, by 2050, solar PV and batteries have become so cheap that wind cluster #8, with its unfavorable class-III CAPEX and hence relatively high LCOE ([Figure S4](#)), is unattractive not only in comparison with other wind clusters but also with solar cluster #2 in combination with batteries (the only active solar cluster that is not yet "maxed out" by 2050).

Thus, as for solar PV, it is shown that the selection preference for wind clusters is *nontrivial*. Although the very highest- CF clusters do tend to be selected first, regardless of remoteness, outside of these "top" resource locations, the relationship between CF , $trCAPEX$, and profile shapes becomes intricate.

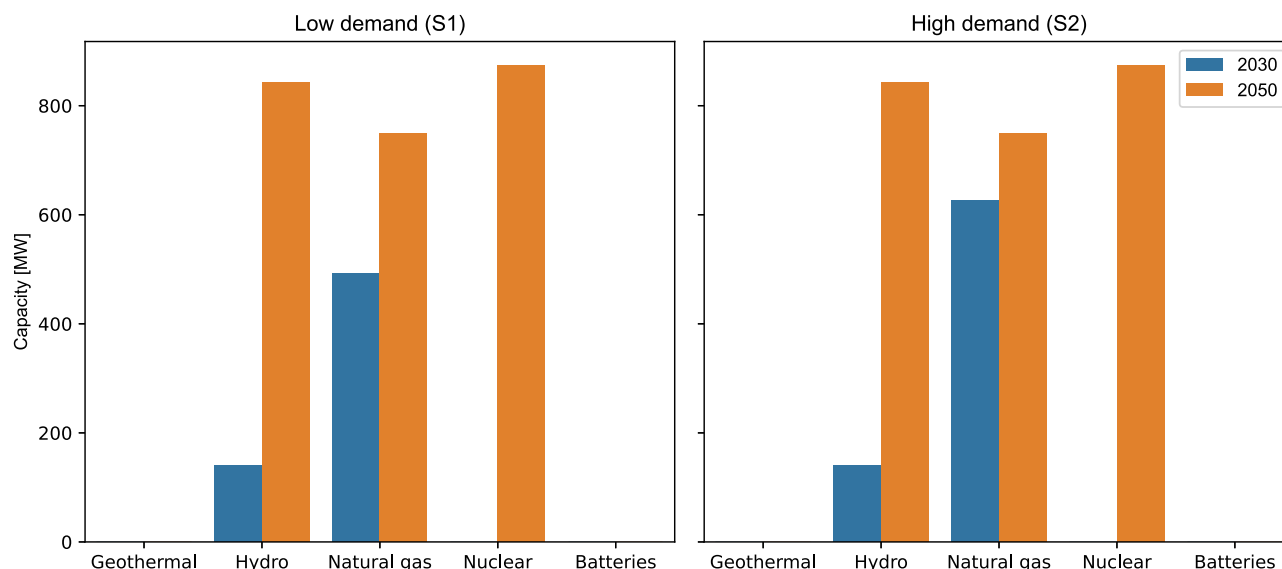


Figure 4. Additional generation capacities identified by FlexTool

Additional generation capacities suggested by FlexTool for 2030 and 2050 on top of the OSeMOSYS results to deal with flexibility challenges at the dispatch level. There is also additional investment in transmission capacities between different regions of the Kenyan grid, which are described in the text but not shown here.

Thus, the added value of the geospatially referenced representation through MSR clusters is again showcased.

The observed trends are robust under less-optimistic assumptions for the future cost reductions of solar and wind power, as shown in the two additional high-cost scenarios that were run. The results show that going to more-pessimistic capital cost projections for VRE does not change the order of VRE cluster selection, and it only delays the onset of large-scale VRE buildout by several years. Results can be consulted in [Figures S1 and S2](#).

Finally, to further underline the added value of the MSR approach (apart from adding layers of geographical information in the results), we provide a comparison of the results of the S1 scenario with a version of the model with the same cost and demand assumptions but without spatial disaggregation (see [Figure S5](#)). This non-geospatially referenced scenario exhibits a much more unbalanced investment in wind power, nearly pushing solar PV out of the system and also strongly reducing investments in geothermal resources, providing less-realistic pathways, with nearly all investment concentrated in a single resource resulting in a highly undiversified electricity mix. Clearly, the MSR-based approach not only provides a spatial disaggregation of otherwise similar results but also provides more-refined pathways overall.

Flexibility analysis

When analyzed in FlexTool, both scenarios show similar results for 2030. The model's optimal dispatch plan identifies only one flexibility challenge, the lack of inertia in the system, which amounts to 0.2% of the requirement for both cases. There are no curtailments in both scenarios. To solve the lack of inertia, the investment mode of the model suggests additional investments in natural gas and hydropower capacity in both cases (see [Figure 4](#)). Notably, no new transmission line investments between the regions of Kenya's power system areas (Western, Nairobi, Mt. Kenya, and Coast) are recommended for 2030.

In the low-demand case, for 2050, FlexTool reveals more-substantial flexibility challenges in the dispatch planning of the power plants. Whereas the core issue remains insufficient inertia (although less than 0.1% of the requirement), minor loss of load (0.07% of demand) and reserve deficiencies (0.6% of demand) also occur. Curtailment of VRE rises considerably to 26%.

FlexTool's investment mode recommends a mix of power capacity additions ([Figure 4](#)) to address these issues (for a more-detailed explanation of what "investment mode" entails, see Kihara et al.³²). Natural gas (0.8 GW), nuclear (0.9 GW), and hydropower (0.7 GW) are introduced to tackle the lack of inertia thanks to their high inertia constants (see [Table S2](#)), and 0.1 GW, corresponding to the High Grand Falls dam, is suggested for its ability to provide flexibility, inertia, and reserves. These investments eliminate flexibility issues and reduce curtailment to 10%, as demonstrated in [Figure 5](#).

The Invest graph (i.e., dispatch, including additional investments suggested by FlexTool) in [Figure 5](#) shows the flexible operation of nuclear and geothermal power plants, with periods lasting several hours of complete shutdown before ramping up again to minimum load or higher. This optimized dispatch plan is the result of the following constraints: a minimum hourly load of 50% and 75% for nuclear and geothermal, respectively; minimum up and down times of 48 and 1 h; and ramp-up and ramp-down rate limitations of 1% and 2%–3% of capacity per min. In this light, further analysis is recommended on the feasibility of the suggested capacity additions because, e.g., the frequent shutdown of nuclear power plants in these results requires more scrutiny.

However, we note that the precise mix of natural gas, geothermal, and nuclear capacity suggested by FlexTool is of secondary importance for the results presented here. The main point is that additional capacity, to the tune of 2–3 GW from units capable of providing baseload and flexibility, such as natural gas units,

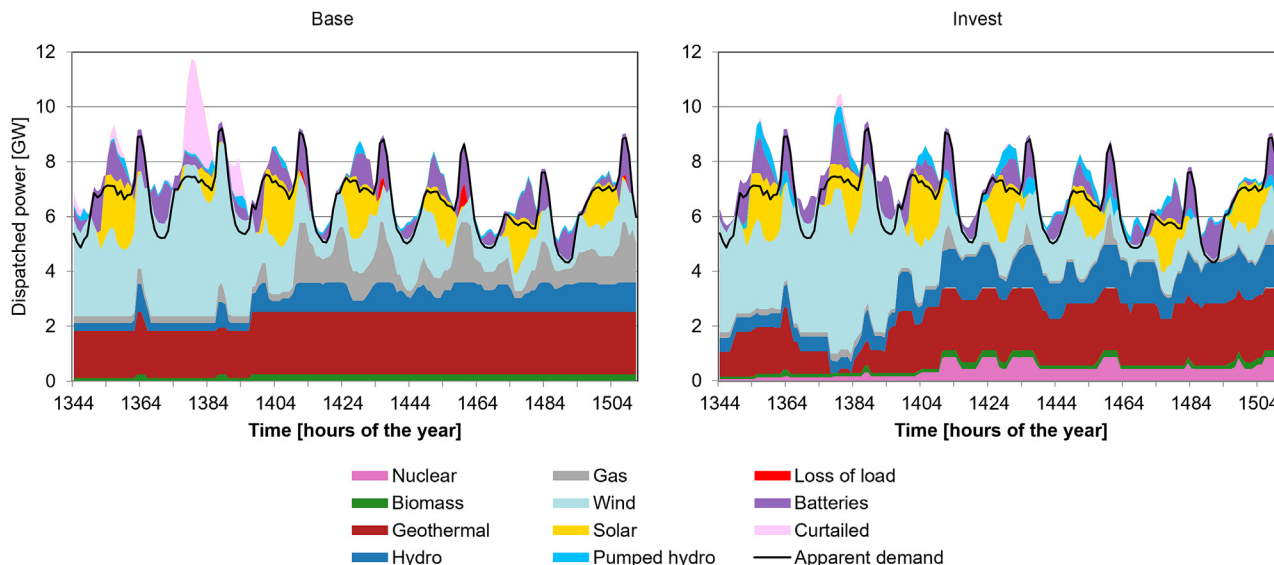


Figure 5. FlexTool dispatch profiles

Dispatch of the power plants in the low-demand scenario for 2050, before (base) and after (invest) investments suggested by FlexTool. The apparent demand represents the domestic demand plus exports minus imports. The 2.5 GW capacity addition in the invest case changes the dispatch profile: nuclear provides a significant contribution to baseload capacity and inertia, allowing geothermal to be run more flexibly. Combined with increased pumped hydro capacities, a higher share of renewables can be integrated in the network, substantially reducing curtailment.

geothermal plants, or nuclear plants, would be necessary to allow the VRE-heavy mix suggested by OSeMOSYS to be realistic.

In the high-demand case, for 2050, there are no flexibility issues, but curtailment of VRE rises to a substantial 32% of generation. However, FlexTool's investment mode still suggests some additional capacities (Figure 4), leading to a lower operational cost of the system: 7.4 billion dollars instead of 8.4 billion dollars before the investments. Curtailment slightly decreases to 31%, thanks to investments in transmission lines. This relatively high amount of remaining curtailment could be potentially reduced by including exports in the model, which is currently not possible with FlexTool. Moreover, the current investment mode does not penalize curtailments, which does not incentivize the model to invest in storage solutions like batteries, which could further reduce curtailment in dispatch mode. Investments in new transmission lines between the Nairobi, Mt. Kenya, and Coast regions amount to 1.2 GW, around 15% of the base-year capacities. The primary focus is on strengthening the Coast-Nairobi line with 0.8 GW, with the remaining on the Mt. Kenya-Coast line with 0.4 GW.

DISCUSSION

This study describes the successful creation of a capacity expansion and dispatch optimization model based on OSeMOSYS and FlexTool, with the explicit inclusion of geospatially differentiated investment options for VRE. The result is a model that not only produces scenarios on *how much* VRE capacity to deploy in a country system in the future but also *where* and *in which spatial sequence* to build this capacity.

Applying the model to a case study for Kenya, the results show that, in terms of site selection, preferences are nontrivial in the sense that they are *not* necessarily determined by the CF of

the resource, which is the result that one would have expected from a classical model that does not have a geospatial differentiation of costs based on infrastructure expansion needs. For both solar PV and wind power, the site selection preference in a cost-optimized portfolio is a function of resource strength, grid expansion needs, and temporal profiles, with none of the three factors decisive by itself, the exception being the very top wind speed locations, where resource strength defaults to being the sole factor of importance.

In the Kenyan case, our scenarios suggest that solar PV and wind power may jointly come to form the backbone of Kenya's future power system. The solar PV and wind resources will be preferentially spread across different parts of the country following the aforementioned factors and will be backed up by geothermal power plants, natural gas power plants, battery storage, and pumped hydro storage. The results also show that generation, transmission, and storage expansion can be co-optimized, and separate planning of the three is to be avoided.

Further, the study shows that, to safeguard system security, VRE capacity in Kenya could be sequentially developed over various locations in a clear priority order determined by various site characteristics, allowing harnessing of the resources not only according to their strength but also according to how they contribute optimally to the system under given constraints. We note that this geographical spread would not only make the system more robust to typical disruptions (e.g., individual plant or line outages or adverse localized meteorological conditions) but also to external shocks such as conflict, as argued by Spyrou et al.⁴³

Although the above is in general agreement with previous work in Kihara et al.,³² various options exist to improve the work, which we touch upon below.

First, with the main focus of this study being the inclusion of the geospatially referenced VRE clusters in the model, going forward, we propose that more research is required to ensure all other technologies are also represented at the appropriate level of detail and ascertain that the suggested mix of technologies in this study is both plausible and practical. This refers principally to the future costs of technologies such as natural gas plants (requiring LNG import infrastructure to be built out in Kenya), pumped hydropower plants and their potential siting, new geothermal projects and their infrastructural needs, etc.

Second, the model used here considered Kenya as a mostly islanded system, with existing interconnections to neighbors such as Ethiopia modeled but no opportunity for the model to invest in other cross-border interconnections. In the future, the model could be upgraded to include these potential links and examine effects of large VRE expansion on power trade with neighboring regions.

Third, it could be argued that the cost of grid expansion for selecting optimal solar and wind clusters should itself be a dynamic function of VRE buildout within the model. For instance, as the first clusters get selected and investments in transmission lines to link them to the existing grid are made, other nearby clusters would technically see their distance-from-grid parameter drop from their initial value. Such a dynamic update of grid extension CAPEX is, however, not considered in the present model, the main argument being as follows. Imagine two clusters, A and B, which both lie along the same path from the nearest existing grid connection point, but B at double the distance from the grid as A. If the model selects cluster A for VRE buildout and grid connection in a certain year, the distance from B to the “existing grid” would, in theory, subsequently halve, the grid having already been built out up to A. But a buildout of VRE in B would still require the same investments in *transmission capacity* as before, across the same distance as before—as the investments made by the model in transmission infrastructure connecting cluster A will only evacuate up to the maximum capacity of cluster A and thus not be sufficient to accommodate cluster B unless more transmission capacity across the entire distance from cluster B to the initial grid is built.

Fourth, at the geospatial level, we further recommend on-the-ground fieldwork for each of the suggested VRE buildout clusters from the model to ascertain the extent to which conditions there may be appropriate for the construction of utility-scale solar PV and wind power projects. This needs to consider the social, cultural, and ecological dimensions of planning potential megaprojects in areas shared with human settlements, agricultural activities, and biodiversity hotspots, or access to which would require traversing terrain with such elements. Although the MSRs approach principally excludes from the outset any protected areas, restricted areas, or areas with unsuitable land use, this is only a first high-level exclusion that may not be able to capture many on-the-ground elements, such as local opposition to megaprojects or wildlife migration corridors impacted by the projects. We comment that, if necessary, it is possible to re-run the analysis leading to the solar and wind cluster creation with stricter exclusion criteria in case this would turn out to be necessary.

Fifth, the workflow presented here only considered on-grid systems, but possibilities may exist to soft-link the present model with geospatially referenced models for off-grid electrification, such as OnSSET. In such a setup, OnSSET would help

to assess the demand that could be met by the grid versus off-grid solutions, whereas the OSeMOSYS-FlexTool workflow assesses the cost of grid electricity, which would in its turn be an explicit input into OnSSET. The fact that OSeMOSYS-FlexTool would itself have an explicit geospatial component in this setup, would help to more-clearly distinguish areas where the grid would arrive in the future from areas that may remain uncovered by utility-scale infrastructure.

Sixth, we stress the importance of not seeing the current model results as a fixed outcome but rather as a snapshot in a type of study that requires periodic updating to keep track with developments on the ground. As various external factors (outside of VRE cluster selection) lead to an expansion of Kenya’s grid infrastructure, which areas are considered “close” and “far” from the existing infrastructure may organically change in the coming decades, with associated implications for the outcomes of site selection priorities by the present model. Thus, a periodic updating of all data inputs is recommended to keep abreast of concurrent developments. To support this, the authors have made the input data for OSeMOSYS and FlexTool used for this study publicly available (see “[resource availability](#)”) and will strive to undertake capacity building with relevant stakeholders in the data and model use.

As stated in the [introduction](#), one of the principal goals of this study was to generate insights to allow governments to set up spatially explicit priority roadmaps to attract investment for VRE plants, with beneficiaries being primarily private investors as well as local and international finance institutions. We believe that these stakeholders may be the primary target groups of the information and approach shown in this study. In this spirit, the author team will next work on setting up data visualization infrastructure to interactively present the geospatial outcomes of the current study, to be embedded within the Energy Access Explorer (www.energyaccessexplorer.org) to enhance integrated and inclusive energy planning at both national and subnational level in Kenya and to inform the upcoming Kenya National Electrification Strategy.

Finally, we emphasize the high replicability of this kind of study to other country contexts. The developed methodology and input datasets are model-agnostic and could be readily applied in other frameworks such as PyPSA. The MSR methodology has already been used to generate results on siting options for solar and wind plants for all of continental Africa and also, more recently, Central and South America,⁴⁴ and, as such, any country-level energy modeling exercise on those continents could readily be extended to include the present geospatial explicitness. We note that the Kenya model developed for this study has low computational requirements, with a single model scenario run being possible on a standard laptop in around 10 min. Given all the above, we hope that further initiatives may be inspired by the present approach in the future.

EXPERIMENTAL PROCEDURES

OSeMOSYS-FlexTool linkage

This paper uses a combined modeling approach to analyze the Kenyan power system by leveraging the strengths of two open-source energy system models: OSeMOSYS and FlexTool.

The OSeMOSYS is a cost-optimization model for long-term energy planning, driven by energy demand.⁴⁵ It relies on a detailed bottom-up description of the energy system, encompassing technologies, resources, and energy carriers. The model uses linear programming to optimize the total cost of the energy system, subject to a set of constraints representing technical or environmental limits. OSeMOSYS leverages a community of practice built around three pillars, namely a code management structure, a community forum, and outreach activities.⁴⁶ The full documentation for the model is available online.⁴⁷

FlexTool is a power system optimization model, developed by IRENA,^{3,48} which uses linear programming to solve the unit commitment and economic dispatch problem. The model provides the least-cost optimization dispatch of power plants with a detailed assessment of potential issues linked to power system flexibility, including curtailment, loss of load, excess load, or insufficient reserves and inertia. Beyond the dispatch problem, FlexTool can also select the least-cost investment options to address the identified flexibility issues. The model assumes perfect foresight, typically using a time resolution of 1 h or less. Additionally, FlexTool is an open-source model and freely accessible for broader use.⁴⁹

OSeMOSYS is particularly suited for long-term planning as it optimizes the energy system for the entire horizon. On the other hand, FlexTool optimizes at an hourly level within a single year, incorporating detailed power system operational constraints not included in OSeMOSYS. The soft-linking between the two models allows FlexTool to complement the OSeMOSYS analysis, which has a much-reduced representation of flexibility.

As described in Kihara et al.,³² the soft-linking process relies on a bidirectional exchange of information. For each chosen marker year (the years for which FlexTool is run based on OSeMOSYS results for that year to identify potential flexibility challenges in that year), FlexTool assesses the flexibility of OSeMOSYS power system capacities and identifies additional power plant or transmission line capacities to solve the flexibility issues. These additional capacities are then fed back as constraints into OSeMOSYS, leading to a revised pathway for the power system capacities' expansion. The process is then repeated for the next marker year. In this study, the chosen marker years were 2030 and 2050.

OSeMOSYS description

For details on the OSeMOSYS setup, the reader is referred to Kihara et al.³² As compared with that publication, four main additional elements were included in OSeMOSYS, which are explored here.

The first is the inclusion of the geo-referenced solar and wind MSRs as separate technologies, replacing the former “generic” solar and wind investment option. Details are given below in “[representation of solar and wind investment options](#).”

The second is the inclusion of buildout speed constraints in each of the separate geo-referenced VRE clusters to ensure *N-1* security adherence,³⁷ avoiding too-high concentrations of VRE in areas evacuated by a single transmission corridor. Again, details are given below in “[representation of solar and wind investment options](#).”

The third adaptation is applied at the level of the time-slice definition and stems from the observation of high penetration of wind power in the results for the Kenyan context.³² Using time slices at seasonal and daily time-scales allows the capture of the main variabilities of demand, hydropower profiles, and solar resources, but it leaves out the substantial fluctuations of wind power yield at the “in-between” resolution of several days to weeks. Using classical seasonal and daily time slices can thus lead to averaging out any effect of prolonged periods with bad wind conditions (wind droughts, sometimes referred to as “*Dunkelflauten*” when concurrent with unfavorable solar conditions), which cannot be captured at a daily or seasonal level but would have strong impacts on system flexibility and reserve needs in practice (how to better represent the possibility of such periods occurring remains an ongoing area of study in capacity expansion modeling^{50–53}).

The result is that wind power often appears to be a more-stable power source in time-sliced results than it actually is, i.e., the variability of the wind resource is under-represented. In some cases, this can lead to wind power almost appearing as “baseload” technology, leading to unrealistic model results. One could

solve this by modeling at full hourly resolution, but this is usually prohibitive computationally in long-term capacity expansion models. One potential way to address this in capacity expansion models without unduly burdening computational requirements is, therefore, to explicitly distinguish periods of low and high resource availability, inspired by, e.g., Helistö et al.,⁵⁰ Nahmmacher et al.,⁵¹ Norvaiša et al.,⁵² and Poncelet et al.⁵³ This allows for the better approximation of the statistical properties of wind across time slices.

Here, inspired by the above, we use a similar principle to better-represent wind variability: adding a fifth “virtual season” to the time slices setup in OSeMOSYS to capture the potential effect of wind droughts⁵⁴ (this approach works as long as the geographical area under consideration is not larger than typical weather systems, in which case it is no longer possible to define such a season for all regions under consideration). This virtual season is defined to cover only 10% of the “worst” wind days of the year, i.e., 10% of the days with the lowest CF on average over all wind clusters weighted equally. This season is thus hypothetical in that it represents a non-chronological set of days. Within this season, no distinction is made between weekdays and weekend days. Correspondingly, all days selected for this season are removed from their original (chronological) season to avoid double-counting (see [Section S1.4](#) and [Tables S3](#) and [S7](#)).

We note that this addition of a virtual weak wind season has a considerable effect on the representation of the wind resource: in this season, the average wind CF is typically only a fraction of its yearly average, representing truly bad wind conditions. A full table with the numbers per cluster is given in the [supplemental information](#) (see [Tables S4](#) and [S8](#)). We conclude that the inclusion of this season works well to explicitly represent the possibility of wind droughts in the Kenyan system.

The fourth adaptation refers to the explicit inclusion of costs for building an LNG terminal to have a more-refined representation of the future costs of expanding natural-gas-based power generation infrastructure. From 2030 onward, it is considered that Kenya could add a maximum annual LNG import capacity of 1 Mton/year every 5 years, hence reaching a total maximum of 5 Mton/year by 2050. The cost of a 1 Mton/year import terminal is estimated to be 570 MUSD, plus an additional investment of around 300 MUSD for the infrastructure needed to bring the regasified natural gas to the consumers (power plants). Such investment costs add up to a capacity cost for natural gas import infrastructure of approximately 29 MUSD/PJ/year.

FlexTool assumptions

The main assumptions in FlexTool are described in Kihara et al.³² The technical and economic parameters of the power plants and transmission lines are aligned with OSeMOSYS. Values for the minimum loads per technology were provided by the LCPDP team, except for the case of geothermal, where the authors assumed a minimum load value of 75% to reflect a preference for baseload operation (see [Table S2](#)).

The contribution factor of solar and wind power to reserve provision was set to 50% of their hourly curtailed production (see [Section S1.3](#) and [Table S1](#)). Reserves refer to capacity that will not be used for electricity generation but that can provide upward operational capacity requirements in case of contingencies occurring in the power system.

Penalty costs associated with the different constraints of FlexTool are shown in [Table S5](#). For each year, the inertia limit is computed using the following formula $\frac{1}{2} * P_{loss} / RoCoF$ with f the nominal frequency of the system (50 Hz for the Kenyan power system), P_{loss} the maximum power that could be lost, and $RoCoF$ the rate of change of frequency. The Kenyan power system has not set a rate of change of frequency, but for this study, a value of 1 Hz/s has been assumed. As the maximum power loss changes based on the marker year, the inertia requirement also varies by year.

In our scenarios, P_{loss} typically corresponds to the total capacity located in a single VRE cluster selected by OSeMOSYS. For instance, in the 2050 low-demand case, the largest potential loss is 1.97 GW for several clusters for both solar and wind. This leads to an inertia constraint of 35.5 GWs. We deliberately adopt a conservative approach to ensure sufficient inertia for system reliability, although we acknowledge that wind power production fluctuates, and the maximum loss might be lower depending on the time slices in which the contingency happens.

Compared with the version described in Kihara et al.,³² FlexTool has been disaggregated into four regions that correspond to the power system areas used by the LCPDP team in their medium and long-term plans^{38,39}: Western, Nairobi, Mt Kenya, and Coast. All current and future power plants are thus associated with one of the four regions. Each solar and wind cluster is thus allocated to one region. For generic capacities of hydro, gas, and batteries, which are not location specific, the following assumptions apply: hydropower potential is assumed to be developed in the Mt Kenya region, gas in the Coast, and batteries are installed for 75% in the Western region near solar and wind farms, with the remaining capacity in Nairobi. Imports are assumed to directly reach Nairobi.

The base-year capacities for inter-regional transmission are provided by the LCPDP team for 2019. In dispatch mode, these capacities for the marker years remain unchanged compared with the base year. In investment mode, the maximum capacities of the different transmission lines increase over time, as shown in Table S6 (see also Section S1.5).

Representation of solar and wind investment options

The geospatially differentiated representation of solar PV and wind power potential was based on the concept of MSRs introduced by IRENA and published in Sterl et al.³⁶ Starting from existing maps on resource strength (irradiation, wind speed, etc.) and on geospatial limitations (conflicting land use, unfavorable slope, protected areas, high population densities, existing road/grid infrastructure, etc.), the MSR workflow allows the identification of each country's feasible regions for utility-scale VRE expansion. Through a statistical downscaling approach applied on reanalysis data, the MSR method combines the benefits of high spatial resolution of VRE potential (kilometer-scale, thus avoiding conflating regions with highly divergent strengths into the same "pixel") with those of high temporal resolution (hourly).

Subsequently, the MSR workflow disaggregates the identified regions into individual units that represent potential individual power plants with maximum sizes ranging from several tens of MW to a few GW (depending on geographical constraints), each with their own CFs, hourly production profiles, and investment costs. Here, the latter reflect not only the costs for the plant itself but also the required transmission line (and corresponding substation) buildout and road network buildout to connect the location to existing grid/road infrastructure. Each resulting individual unit is called an MSR, the smallest spatial unit that has unique techno-economic characteristics.

In principle, MSRs can be directly input into capacity expansion modeling, as the required techno-economic metadata are available and unique for each. However, there are usually thousands of such regions available at the country level whose overall potential vastly outnumbers the projected future demand, potentially leading to overcomplicated models with high runtimes. Therefore, Sterl et al.³⁶ recommend two further steps to keep MSR-based models "light" in size.

The first step consists of ranking the MSRs according to a certain "attractiveness" criterion, such as the expected LCOE at the power plant level, and only including the most attractive segment in the capacity expansion model. For instance, Sterl et al.³⁶ recommended including only the top 5% (by LCOE, including transmission and road network expansion costs in the plant CAPEX) in terms of country area coverage.

The second step consists of using a mathematical *k*-means clustering method to group the remaining top MSRs (still typically up to several hundred) into a limited number of "clusters," each characterized by extremely similar shapes of the temporal production curves (given the typical decorrelation distance of VRE resources, this approach tends to create clusters made up of spatially contiguous MSRs, though in some cases a single cluster could consist of different "islands" of very high and similar potential separated by areas of lower resource strength). The idea is that, in terms of system planning, it would not matter a lot where exactly within the cluster future power plants would be built, as the individual locations within the cluster are of highly similar nature. Sterl et al.³⁶ recommended grouping the MSRs into around 10 clusters for countries of a size like Kenya's.

In this study, we followed the recommendations by Sterl et al.,³⁶ including 10 clusters of solar PV MSRs and 10 clusters of wind MSRs as individual tech-

nologies in the OSeMOSYS-FlexTool workflow, each with their own unique profiles, investment costs, and maximum deployment potential. The original datasets are available on Zenodo.⁴¹ We also followed the recommendations by Sterl et al.³⁶ in allocating loss factors between the hourly profiles that the VRE plant produces and what eventually reaches the grid of 8% for solar PV (outage and inverter losses) and 17% for wind (outage, array, and wake losses).

The clusters in Sterl et al.³⁶ were developed based on an assumed wind turbine hub height of 100 m. Because Kenya's largest wind farm to date, the Lake Turkana Wind Power Project, used somewhat lower-hub wind turbines, we calibrated the cluster-level wind power generation profiles to LCPDP data with a constant ratio, represented by the ratio of average CF between the LCPDP-delivered time series for Lake Turkana Wind Power Project on one hand and the wind MSR #4 (in which the Lake Turkana Wind Power Project falls) on the other. This ratio turned out to be 0.85.

After having included the clusters and their profiles in the model workflow, using the aforementioned grid and road expansion CAPEX to distinguish the clusters from each other at the cost level, the model required further adaptations to realistically represent power system expansion, taking into account system security constraints. In particular, the model should be prevented from indiscriminately accumulating VRE capacity within the same cluster. The reason is that a high concentration of power generation capacity within the same geospatial zone, using (at least partly) the same evacuation infrastructure, would likely violate *N-1* constraints, as argued, e.g., by Donk et al.³⁷ From a system security point of view, all capacity within a single cluster, assumed to be evacuated by the same transmission line, acts as a single power plant (even if, in reality, it could be made up of several different projects), and the potential loss of a single power plant should never exceed the reserve margin of the system for *N-1* security. Hence, the total allowed capacity within each cluster in any modeling year was constrained to a maximum corresponding to the reserve margin of the system (in MW) in that year (so long as that number did not exceed the maximum theoretical deployment potential in the cluster, in which case the latter number prevailed as the overall ceiling). The reserve margin, in its turn, was set to 10% of the peak demand in each year.

Scenarios

Four scenarios have been modeled in this work, based on two different demand projections (referenced in this text as low and high demand) and two capital cost projections for solar and wind (again referenced in this text as low and high cost). The two demand projections considered are the ones used in Kihara et al.³² and are based on the latest version of the LCPDP, published in 2022.⁵⁵ More details on demand profiles and total annual demand projections are given in Section S1.1. Solar and wind capital costs are based on projected cost reductions taken from the IEA's World Energy Outlook (WEO).⁵⁶ The low-cost scenario corresponds to the IEA's net zero (NZ) scenario, whereas the high-cost scenario corresponds to the stated policy (SP) one. A complete description of how capital costs have been obtained is given in Section S1.2. The resulting four scenarios are hence the following:

- S1: low demand, low solar, and wind capital costs projections.
- S2: high demand, low solar, and wind capital costs projections.
- S3: low demand, high solar, and wind capital costs projections.
- S4: high demand, high solar, and wind capital costs projections.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Pietro Lubello (p.lubello@ucl.ac.uk).

Materials availability

This study did not generate new, unique materials.

Data and code availability

The OSeMOSYS model and data files used in this study can be found on the GitHub repository at: https://github.com/ClimateCompatibleGrowth/osemosys_kenya_psm, and the specific version of the model was released as v2.0.0 on Zenodo: <https://doi.org/10.5281/zenodo.11165893>. The Kenyan FlexTool model is available online on Zenodo: <https://doi.org/10.5281/zenodo.11165550>, and the model code (FlexTool v2.0) can be downloaded at <https://www.irena.org/Energy-Transition/Planning/Flextool>.

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AUTHOR CONTRIBUTIONS

A.M. and P.L. carried out the modeling work with OSeMOSYS and FlexTool under the guidance of S.S., who conceived the study. E.M.T., M.M., and M.A. supported calibration of the models to the Kenyan context. All authors contributed to the drafting and reviewing of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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